
Task #4 - Association rules

Association Rules Mining is a **pattern discovery** technique used on large datasets to identify meaningful relationships between variables. It is commonly applied in **transaction analysis** (Market Basket Analysis) to determine which products are frequently purchased together.

An **association rule** is an expression of form:

$$X \Rightarrow Y$$

where:

- **X** (antecedent) → A set of items that appear together in the database.
- **Y** (consequent of the rule) → Item(s) that are often associated with **X**.

Used metrics

To evaluate the association rules, the following indicators are used:

1. Support

Support for a set of items:

$$\text{Support}(X) = \frac{\text{number of transactions containing } X}{\text{total number of transactions}}$$

Support for an association rule:

$$\text{Support}(X \Rightarrow Y) = \text{Support}(X \cup Y) = \frac{\text{number of transactions containing } X \cup Y}{\text{total number of transactions}}$$

2. Confidence

$$\text{Confidence}(X \Rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)} = \frac{\text{number of transactions containing } X \cup Y}{\text{number of transactions containing } X}$$

Confidence expresses the probability of buying **Y** when **X** is bought. A confidence of 80% indicates that **80% of the clients buying X also buy Y**.

3. Lift

$$Lift(X \Rightarrow Y) = \frac{Support(X \cup Y)}{Support(X) \cdot Support(Y)}$$

Interpretation:

- **Lift > 1** → the occurrence of X **increases** the probability of Y
- **Lift = 1** → X and Y are independent
- **Lift < 1** → the occurrence of X **decreases** the probability of Y

Extracting association rules

1. Frequent itemsets are identified, meaning those sets whose support exceeds a fixed threshold.
2. Association rules are identified, built on the elements of the frequent itemsets, which exceed a fixed level of confidence.

Apriori algorithm - the most well-known algorithm for discovering **frequent itemsets**. It uses a **pruning**-based strategy (eliminating unlikely combinations) to make the search efficient, based on the **Apriori principle**:

Any subset of a frequent itemset must also be frequent.

Main steps: candidate generation, candidate pruning, computing support to keep good candidates. Finally, association rules are generated using frequent itemsets.

Task

1. Discretize the numerical attributes; use multiple methods in order to determine the most suitable method for each feature.
2. Set various thresholds for the *support* and *confidence* metrics.
3. Extract association rules using the Apriori algorithm (optionally also FP-Growth); order the rules by the discussed metrics and identify rules that can have a useful interpretation.
4. Analyze:
 - a. Which data discretization method is preferable ? (2p)

- b. How does the set of rules vary when changing the thresholds from point 2 ? (2p)
- c. Extract useful rules and provide a detailed interpretation of them, and write these interpretations as Colab text cells. (2p)
- d. What rules can be extracted to build a classifier? Write comments in Colab text cells on their quality in terms of accuracy and coverage. (2p)
- e. Create plots to support the analysis of the results (scatterplots on pairs of analyzed metrics and alluvial plots). (2p)